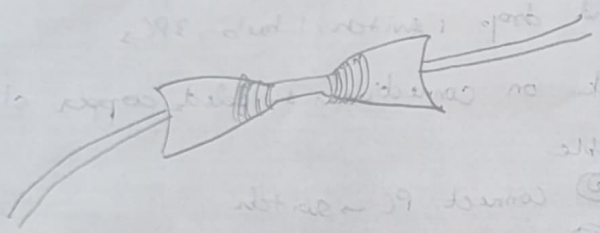
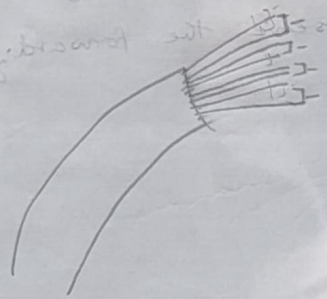
Step1 : To start construction of device, begin by threading shields onto the cable

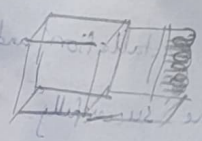


Step2: Next strip 1.5 cm of cable from both ends. The crimping tools fixes a round area to complete this task

Step3 : After you will need to untangle the wires. One end should be in arrangement A, and the other in B.



Step 4: Once the order is correct, bunch them together a line and if there are that stick out farther other snip them back to create an even level.



Ex.No: 2
23/07/2026

NETWORK CABLES

AIM:

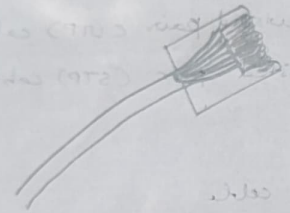
Study of different type of Network Cable

(a) Understand different types of networks cable

1. Unshielded Twisted pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. Coaxial Cable
4. Fibre optical cable

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSION	ADVANTAGE / DISADVANTAGE	APPLICANT USE
UTP	Category 3	10 bps	Advantages: ⊕ change in cost ⊕ Easy to install Disadvantages: ⊖ more prone to EMI & noise	1. 10 Base-T Ethernet 2. Fast Ethernet, Gigabit Ethernet ~~3. Fast ETP~~
	Category 5	upto 100 Mbps		
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages: ⊕ shielded ⊕ Faster than UTP ⊕ Lesser susceptible to noise & interfaces Disadvantages: ⊖ Expensive ⊖ Greater ⊖ Installation effort	Gigabit, Ethernet 10G ethernet widely used in data centers. ⊕ Gigabit Ethernet. 10G ethernet
SSTP	Category 7			
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: ⊕ Higher bandwidth ⊕ Immune to interfa Disadvantages: ⊖ cost ⊖ Size is bulky	Speed of signal is 500 m televisi network high speed inte

Step 5: Next push the cable rigid in, The notch at the end of the plug need to be just over the cable shielding and if it isn't that's means that you're stripped off too much



Step 6: After the wires are securely sitting inside plug.

Step 7: Lastly repeat for the other end using diagram B / using diagram A

Cabel Type	Category	Maximum data transfer	Advantages / Disadvantages	Maximum distance
Fibre optic cable	Single Mode Multimode	100 Gbps	Advantage: ① High speed ② High bandwidth Disadvantage: ① Expensive ② Required skilled installers	Fibre optic cable is around 100 meter

⑤ Make your own Ethernet Cross over / Straight Cable

Tools & parts needed

① Ethernet cabling CAT5e is certified for gigabit support but CAT5 cabling works

② 2 RJ45 plugs

③ Optical 2 plug shields

Result:

Thus the different types of sub network cables are shielded successfully